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Accounting for External Conditions in the Modeling of a Self-Powered Neutron Detector

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Анотація

Назар НЕМОЩЕНКОВ Врахування зовнішніх умов в моделюванні детектора прямого заряду.

Кваліфікаційна робота магістра за спеціальністю 104 Фізика та астрономія, освітня програма «Ядерна енергетика». – Київський національний університет імені Тараса Шевченка, фізичний факультет, кафедра ядерної фізики та високих енергій. – Київ – 2026.

Науковий керівник: доктор технічних наук, Володимир БОРИСЕНКО.

У роботі методом Монте-Карло досліджено вплив параметрів активної зони реактора на сигнал детектора прямого заряду з родієвим емітером. Розглянуто залежність сигналу від концентрації борної кислоти, густини теплоносія та вигорання палива. Показано, що їхня зміна може спричиняти відхилення сигналу ДПЗ близько 20%, що обґрунтовує необхідність урахування цих чинників під час використання ДПЗ.

Ключові слова: детектор прямого заряду, ТВЗ-W, ВВЕР-1000, метод Монте-Карло, вигорання ядерного палива, борна кислота, густина теплоносія.

Summary

Nazar Nemoshchenkov Accounting for external conditions in the modeling of a Self-powered neutron detector.

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The influence of reactor core parameters on the signal of a rhodium self-powered neutron detector was studied using the Monte Carlo method. The dependence of the signal on boric acid concentration, coolant density, and fuel burnup was analyzed. It was shown that variations in these parameters may cause an SPND signal deviation of about 20%, demonstrating the need to account for these factors when using SPNDs.

Keywords: self-powered neutron detector, FA-W, VVER-1000, Monte Carlo method, nuclear fuel burnup, boric acid, coolant density.

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Introduction

Improving the model of a self-powered neutron detector (SPND) is an important task for increasing the accuracy of in-core monitoring, since it aims to determine the neutron flux more precisely and to reproduce the spatial distribution of the power-release field in the reactor core more reliably. It is important to account for operational factors, in particular nuclear fuel burnup, changes in boric acid concentration, and coolant inhomogeneities associated with the temperature gradient along the core height, which lead to corresponding variations in density.[1]

Higher measurement accuracy makes it possible to identify local deviations in the power distribution more reliably and to assess the current state of the reactor core more correctly. This contributes to a more justified selection of operating modes, lower operational risks, more efficient use of nuclear fuel, and an extension of the service life of NPP equipment.

Object of the study: the influence of nuclear fuel burnup, boric acid concentration, and spatial variations in coolant density on the signal of a self-powered neutron detector.

Subject of the study: changes in the sensitivity of a self-powered neutron detector under nuclear reactor operating conditions.

1 Monitoring of Neutronic Characteristics of a Nuclear Reactor Core

1.1 Methods and Means for Monitoring Reactor Parameters

Safe operation of nuclear reactors requires monitoring of power and criticality over a wide range of operating conditions. Reactor power can be monitored by measuring thermal-hydraulic parameters, such as coolant temperature and flow rate, steam flow rate, and related quantities. However, such methods are inertial and do not provide sufficiently prompt monitoring.

At low power, the reactor state is determined by the effective neutron multiplication factor k_{eff} , whereas thermal power cannot be reliably estimated from thermal-hydraulic parameters, since these are governed by residual heat release and depend only weakly on k_{eff} . Therefore, the reactor state is monitored directly through the characteristics of its radiation field.

Reactor radiation is multicomponent and includes fission fragments, neutrons, γ quanta, α and β particles, as well as neutrinos. α and β particles have short ranges in matter, on the order of millimeters for α particles and centimeters for β particles; therefore, their detection characterizes only local processes in the region where they are produced. A significant fraction of β particles and γ radiation is generated by the radioactive decay of fission products and is characterized by an inertial response. Neutrino flux determination is not technically implemented in monitoring systems. Thus, neutrons constitute an important component of reactor radiation suitable for prompt monitoring of the reactor state.

In the first industrial reactors, monitoring was based mainly on measurements of thermal-hydraulic parameters and had limited response speed. Modern monitoring systems are integrated and include measurements of neutron flux, γ radiation, and thermal-hydraulic parameters, thereby improving both accuracy and promptness of monitoring.

Monitoring the state of a nuclear reactor requires measurement of the neutron field over a wide power range. Neutron flux density is the main characteristic used to assess the neutronic state of the reactor and the spatial distribution of energy release.

1.1.1 Ionization Chambers

An ionization chamber is a gas-filled detector in which the signal is formed as a result of gas ionization by charged particles. A potential difference is applied between two electrodes in the sensitive volume. Under the action of ionizing radiation, electron-ion pairs are produced in the gas; they drift toward the corresponding electrodes and generate an electric current in the external circuit. Depending on

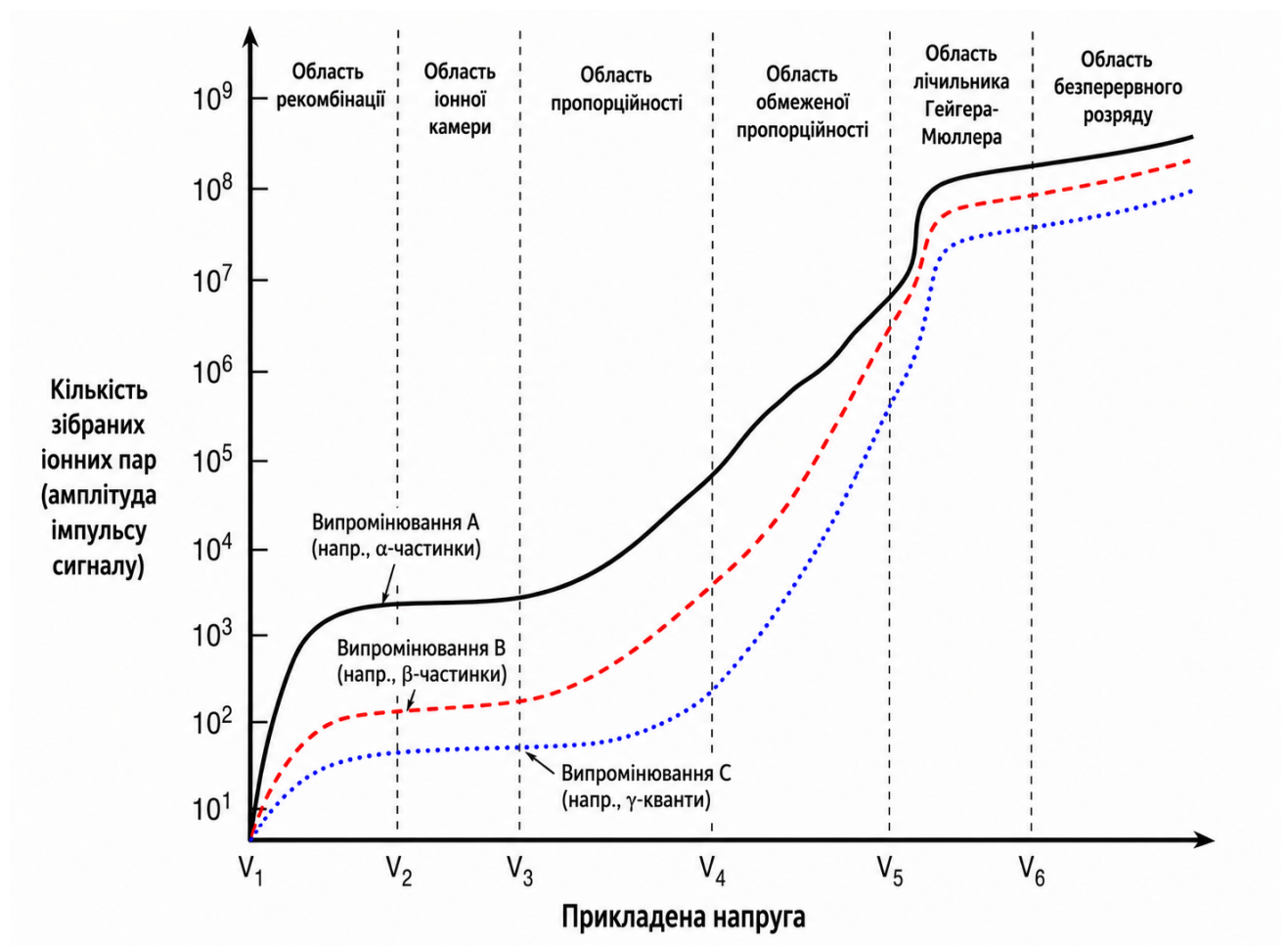


Рис. 1: Operating modes of an ionization chamber

the applied voltage, the chamber can operate in different regimes:

the recombination region, saturation region, proportional region, and others. For neutron detectors, it is important that neutrons themselves do not ionize the gas directly; therefore, their detection requires conversion of neutron interactions into charged particles.

1.1.2 Fission Counters

Fission counters operate on the basis of the fission reaction (n, f) . The detector cathode is coated with a thin uranium layer enriched in the isotope ^{235}U . Under neutron irradiation, uranium nuclei undergo fission, and the resulting fission fragments ionize the gas in the sensitive volume.

1.1.3 Proportional BF_3 Counters

In BF_3 counters [2], neutrons are detected through their reaction with the isotope ^{10}B in the gas volume:



The resulting charged particles ionize the gas, while the detector operates in the proportional region with gas amplification of the signal.

1.1.4 ${}^3\text{He}$ Counters

Helium-3 counters are gas-filled proportional neutron detectors. Their operating principle is based on neutron absorption by a ${}^3\text{He}$ nucleus:



As a result of the reaction, a proton and tritium nucleus are produced. Since both are electrically charged, they ionize the gas in the sensitive volume of the detector. The resulting electron-ion pairs are collected on the electrodes under the applied voltage, thereby forming an electrical pulse.

1.1.5 Linear ^{10}B Counters

In linear ^{10}B counters, boron is deposited as a thin layer on the inner surface of the electrodes. Neutrons interact with the boron layer analogously to reaction (1), and the generated charged particles enter the gas volume and produce ionization.

1.1.6 Activation Methods: SPNDs

Self-powered neutron detectors (SPNDs)[3,4,5,6,7] are used to monitor the distribution of energy release in the cores of nuclear reactors. They are designed to operate under conditions of high neutron flux density, $10^{13} - 10^{15} \text{ n}/(\text{cm}^2 \cdot \text{s})$, elevated temperatures, and a significant γ -background level.

SPNDs have a coaxial structure consisting of an emitter, an insulator, and a collector (Fig. 2). For monitoring energy release in high-power reactors, β -emitting SPNDs are predominantly used; in such detectors, the signal is formed as a result of the production and decay of radioactive emitter nuclei.

Under neutron irradiation, a radiative capture reaction occurs in the emitter material, producing a radioactive isotope:

$$(A, Z) + n \rightarrow (A + 1, Z)^* \rightarrow (A + 1, Z) + \gamma \quad (3)$$

The subsequent β^- decay of the produced nucleus is described by:

$$(A + 1, Z) \xrightarrow{T_{1/2}} (A + 1, Z + 1) + \beta^- + \bar{\nu}_e. \quad (4)$$

In a stationary neutron field, after a time exceeding several half-lives $T_{1/2}$, equilibrium is established in the emitter material between the production rate of

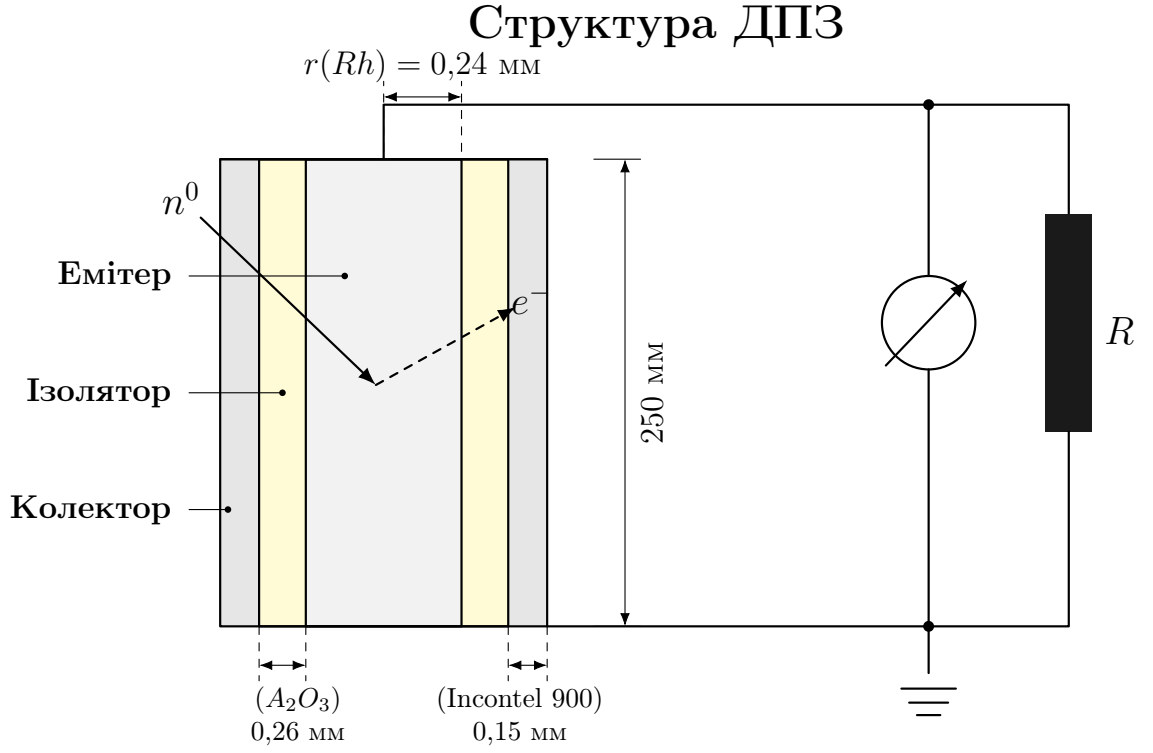


Рис. 2: Structure of a self-powered neutron detector

radioactive nuclei and their decay rate. In this case, the emission rate of β particles is determined by the neutron absorption rate in the emitter.

For an elementary emitter volume ΔV , the β -particle current can be written as:

$$\Delta I_\beta = en\sigma\Phi\Delta V, \quad (5)$$

where e is the electron charge; n is the concentration of emitter nuclei; σ is the microscopic cross section of the radiative capture reaction; Φ is the neutron flux density; and ΔV is the elementary emitter volume. This expression shows that, for a constant emitter composition, the detector current depends linearly on the neutron flux density.

Some of the β particles emitted from the emitter reach the collector and generate an electric current. The materials of the insulator and collector are selected so that their neutron activation is much lower than the activation of the emitter material. In addition to processes occurring directly in the detector materials, the

SPND signal is also affected by the external operating conditions in the reactor core. These conditions include coolant parameters, in particular its density, which changes because of the temperature gradient along the core height; boric acid concentration; and changes in the fuel isotopic composition due to burnup. These parameters modify the local neutron field in the detector region and, consequently, affect the number of reactions in the emitter material and thus the magnitude of the output current.

Various emitter materials are used in self-powered neutron detectors, including rhodium, vanadium, silver, and cobalt. In VVER-1000 reactors at Ukrainian NPPs, SPNDs with rhodium emitters are used predominantly. This is due to the large cross section of the $^{103}\text{Rh}(n, \gamma)$ reaction, which provides high detector sensitivity to the neutron flux. At the same time, rhodium SPNDs also have disadvantages, in particular a higher material cost and more pronounced emitter burnup during reactor operation, which leads to a time-dependent change in detector sensitivity. The transformation scheme is shown in Fig. 3.

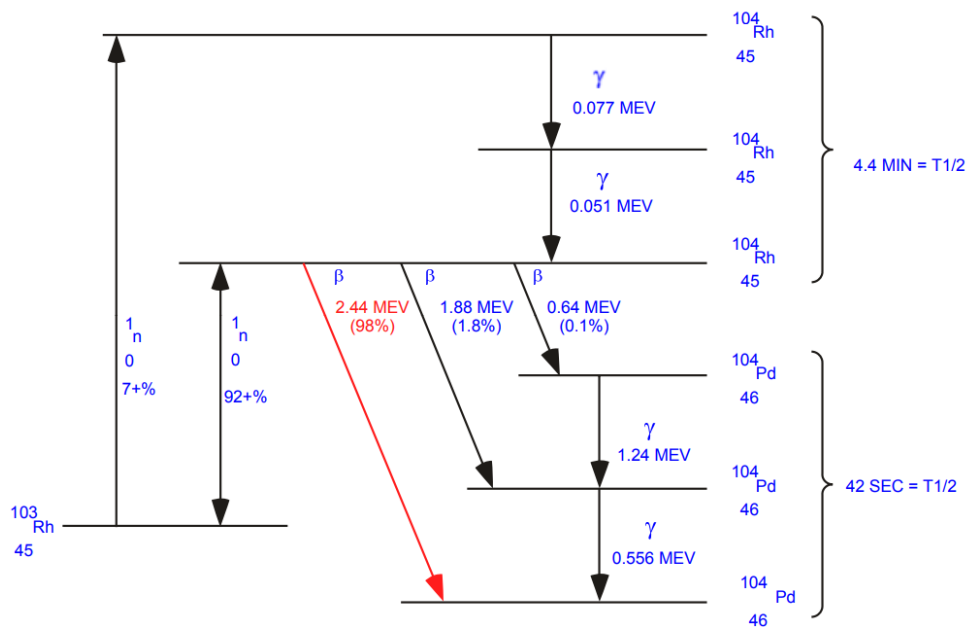


Рис. 3: Rhodium activation and decay scheme in an SPND

2 VVER In-Core Monitoring System

The generalized structure of the in-core monitoring system (ICMS) [8] includes sensors, communication lines, information and measurement equipment, and a computing system. Sensors convert primary physical parameters (temperature, pressure, neutron flux density, etc.) into electrical signals, which are transmitted through communication lines to the information and measurement equipment.

The information and measurement equipment performs measurement, preliminary processing, and transmission of information to the operator and to the computing system. The computing system calculates thermal-hydraulic and neutronic parameters that characterize the state of the reactor core and the reactor facility as a whole. The operation of the equipment and the computing system is carried out according to algorithms and programs, the combination of which forms the mathematical and software support of the system, as shown in Fig. 4.



Рис. 4: Block diagram of the in-core monitoring system operation

In addition to signals from in-core sensors, the ICMS of VVER reactors receives a significant number of signals from ex-core sensors of the primary and secondary circulation loops. This information is used to calculate generalized parameters characterizing the state of the reactor core. The joint use of ICMS data, the control and protection system (CPS), and the neutron flux monitoring equipment provides monitoring of the main parameters of the reactor facility: thermal power, coolant

flow rate through the core and individual loops, temperature regimes, and others.

Thus, in terms of the scope of monitored and calculated parameters, the ICMS effectively performs the functions of a system for monitoring the state of the reactor facility. Although the ICMS is an information system, its failure for a period exceeding allowable limits may require a reduction in unit power or a reactor shutdown.

Temperature and energy-release monitoring sensors are located inside the cores of VVER reactors, as shown in Fig. 5. They operate under severe conditions: high temperature, high pressure, and intense neutron and γ radiation. Therefore, such sensors must have high metrological and reliability characteristics, small dimensions, and structural compatibility with the reactor internals.

Rhodium-emitter self-powered neutron detectors are used in the ICMS to measure the volumetric distribution of energy release in the core. Compared with other types of neutron-sensitive detectors, SPNDs have the following advantages:

1. small dimensions, which make it possible to install a large number of detectors in the reactor and obtain a detailed picture of the spatial distribution of energy release in the core;
2. no need for an external power supply, high reliability, and the ability to operate for at least one fuel campaign;
3. a change in sensitivity during operation that can be accounted for computationally;
4. a simple design, manufacturability, good reproducibility of parameters, and low cost.

At the same time, SPNDs also have certain disadvantages:

1. a small output signal: in VVER reactors at nominal power, the SPND output current is on the order of a few microamperes;
2. a relatively large time constant, which for rhodium SPNDs is about one minute;
3. dependence of sensitivity on reactor core parameters, in particular fuel burnup, enrichment of the nearest fuel rods, boric acid concentration, coolant temperature and density, as well as burnup of the detector emitter itself.

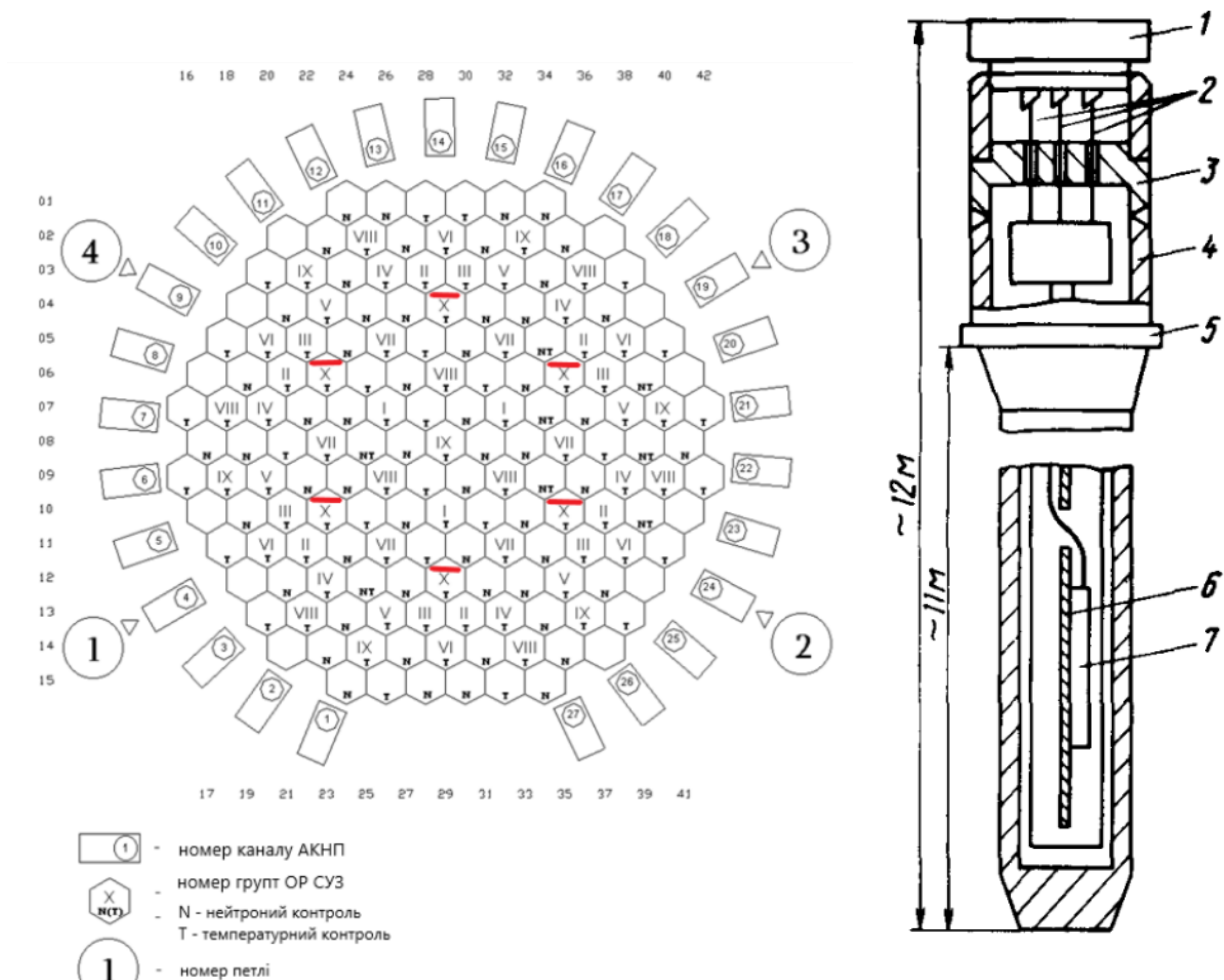


Рис. 5: Arrangement of neutron measurement channels and control rods in the VVER-1000 core

The detector part of a neutron measurement channel (NMC) includes seven DPZ-1M detectors uniformly distributed along the core height with a spacing of

437,5 mm. In addition to the detectors, the channel contains a protective screen, a sealing unit, and communication lines made of KTMS cable. Each communication line, in addition to the signal conductor, includes a background conductor intended to compensate for the current generated in the cable under the action of reactor radiation.

The main part of the SPND output signal is the activation component, which for rhodium detectors reaches approximately 93%.[7] It is formed as a result of neutron activation of rhodium nuclei followed by β decay of the produced radioactive nuclei. Part of the signal is caused by electrons generated on the emitter under the action of γ radiation due to the photoelectric effect and the Compton effect. Both external γ quanta and γ quanta produced during rhodium activation contribute to the formation of this component.

The prompt component associated with the (n, γ) reaction in the emitter usually accounts for about 7–8% of the activation component and is practically inertia-free. Another signal component arises in the communication line under the action of reactor radiation. This background component is proportional to the cable length located in the core and, for a DPZ-1M detector, can reach 8% of the total output signal. To compensate for it, the cable includes a background conductor whose signal is subtracted in the measuring equipment from the signal of the main conductor.

Thus, the expediency of accounting for the influence of reactor core parameters on the SPND signal is determined by the magnitude of the induced changes. These changes must be sufficiently pronounced to be separated from other factors, in particular the contribution of reactor radiation in the communication line, emitter burnup, and other components that also affect the detector output signal. Therefore, the influence of fuel burnup, boric acid concentration, and coolant density is of practical interest only if the resulting change in the signal is comparable to or greater than background and operational contributions.

The SPND output signal is proportional to the local neutron flux density at the

detector location. Since the neutron flux is related to energy release in the nearest fuel rods, SPND signals are used to reconstruct the spatial distribution of energy release in the core, as shown in Fig. 6.

The reactor thermal power based on SPND signals can be determined from the expression:

$$N_{\text{SPND}} = A^{\text{SPND}} \frac{\sum_{i=1}^{64} \sum_{j=1}^7 q_{ij}}{448 - m}, \quad (6)$$

where m is the number of rejected SPNDs for which q_{ij} is taken as zero; A^{SPND} is the conversion coefficient, which is periodically refined in the computing system; and q_{ij} is the signal of the j -th detector in the i -th neutron measurement channel.

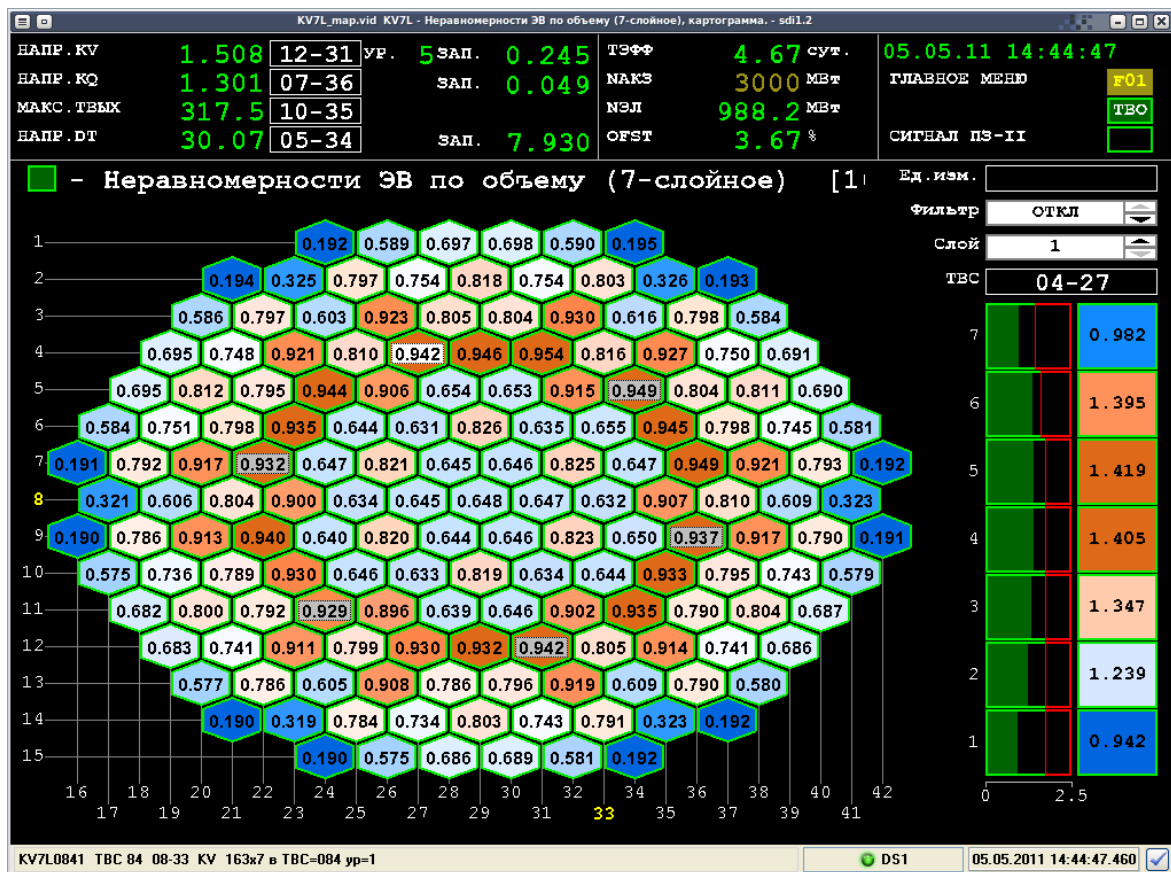


Рис. 6: Example of the energy-release distribution displayed on the in-core monitoring system screen [9].

3 Computational Model of SPND Signal Formation

The MCNP code [10] was selected for modeling the formation of the self-powered neutron detector signal, since it enables detailed simulation of neutron, photon, and electron transport in complex three-dimensional geometries. In addition, MCNP provides advanced tools for specifying materials, nuclear data libraries, and calculation functionals, which makes it possible to estimate reaction rates in individual elements of the model.

3.1 Monte Carlo N-Particle Code

MCNP (Monte Carlo N-Particle) is a general-purpose computational code developed at Los Alamos National Laboratory (LANL, USA). The code is intended for Monte Carlo simulation of neutron, photon, and electron transport.

The Monte Carlo method is based on the statistical simulation of individual particle histories. For each particle, the processes of birth, motion, interaction with matter, and absorption or escape from the model boundaries are reproduced. With a sufficient number of particle histories and a correct specification of geometry, materials, and nuclear data, this approach makes it possible to obtain estimates of neutronic characteristics with a prescribed statistical uncertainty.

In this work, MCNP is used to estimate the rate of the neutron radiative capture reaction in the rhodium emitter of the SPND. This approach makes it possible to relate the local neutron field in the detector region to the signal generated as a result of emitter material activation.

3.2 Description of the Computational Model

The computational model includes a self-powered neutron detector placed in a measurement channel, neighboring fuel rods, and the surrounding coolant. Such

a geometry makes it possible to account for the local conditions of neutron field formation near the detector and to estimate the number of (n, γ) reactions in the emitter material.

For the calculations, a geometric model of a fragment of one FA-W assembly containing a self-powered neutron detector was constructed. The model was implemented in the MCNP, MCNPX, and OpenMC codes. The general view of the computational geometry is shown in Fig. 7.

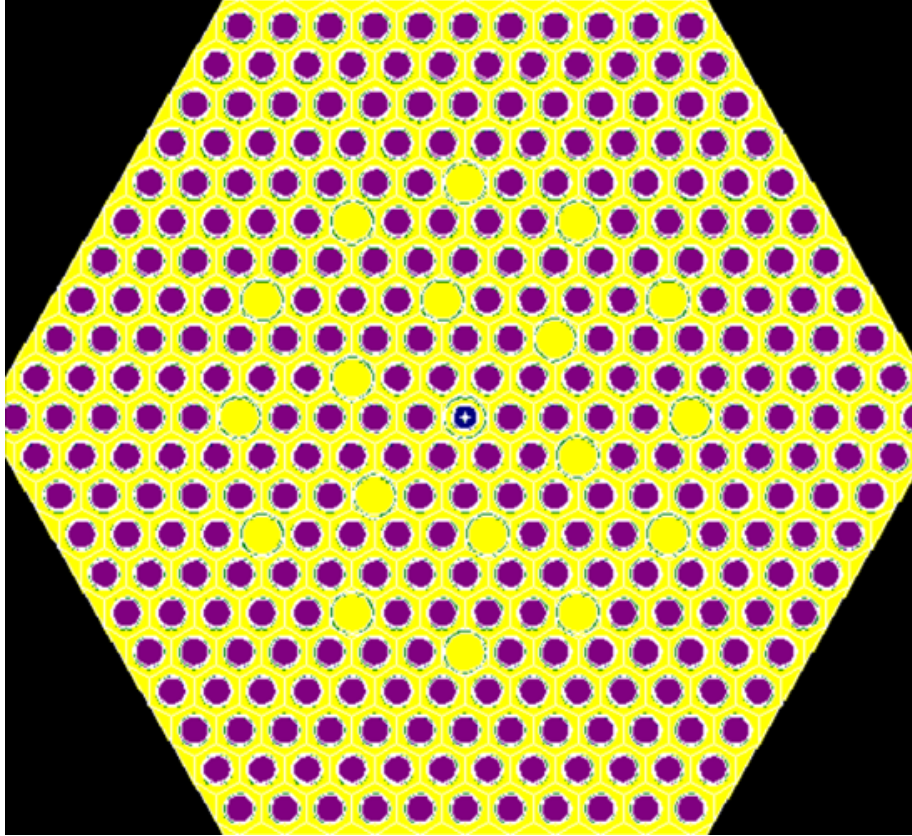


Рис. 7: Geometry of the FA-W computational model with a self-powered neutron detector.

As an initial approximation for the neutron source, the Watt spectrum was used.

$$p(E) = e^{-E/a} \sinh(\sqrt{bE}), \quad (7)$$

The calculations used nuclear reaction cross sections from the ENDF/B-VIII.1 evaluated nuclear data library[11], prepared for a temperature of 600 K.

The number of reactions (reaction rate) in a specified emitter volume is defined as

$$R = VC \int \Phi(E) \sigma(E) dE, \quad (8)$$

where R is the number of reactions in the computational volume; V is the emitter volume; C is the atomic density of ^{103}Rh nuclei; $\Phi(E)$ is the energy distribution of the neutron flux density; and $\sigma(E)$ is the microscopic reaction cross section.

In MCNP, this quantity can be calculated using a tally with an FM card, which specifies a multiplier for the corresponding microscopic reaction cross section. In this case, the tally result corresponds to the reaction rate in the emitter material and can be used as a computational estimate of the SPND signal.

3.3 Estimation of the Detector Signal

For a preliminary estimate of the detector signal, a simplified calculation of the number of reactions in the rhodium emitter was performed. The thermal power of a VVER-1000 reactor is taken as $P = 3000$ MW. With the average energy released per fission event equal to $E_f = 200$ MeV, the fission rate in the core is $F_{AZ} \approx 9,6 \cdot 10^{19} \text{ s}^{-1}$.

The core geometric parameters were assumed as follows: fuel column height $H_{AZ} = 350$ cm, fuel pellet diameter $d = 0,784$ cm. The core contains 163 fuel assemblies with 312 fuel rods in each. Under these conditions, the total fuel volume is $V_{AZ} \approx 8,6 \cdot 10^6 \text{ cm}^3$.

The average fission density in the core is then

$$q_f = \frac{F_{AZ}}{V_{AZ}} \approx 1,1 \cdot 10^{13} \text{ s}^{-1} \cdot \text{cm}^{-3}. \quad (9)$$

For the estimate, microscopic cross sections at a neutron energy of $E = 0,0253$ eV were used:

$$\sigma_f(^{235}\text{U}) = 582,6 \text{ barn}, \quad \sigma_a(^{103}\text{Rh}) = 145 \text{ barn}. \quad (10)$$

The volume of the rhodium emitter in Fig. 2 is $V_{Rh} \approx 0,045 \text{ cm}^3$.

Within the simplified estimate, the number of reactions in the rhodium emitter was determined as

$$R_{Rh} = q_f V_{Rh} \frac{\sigma_{Rh}}{\sigma_f} \approx 1,3 \cdot 10^{11} \text{ s}^{-1}.$$

For comparison with the results of Monte Carlo simulations, it is necessary to account for the fact that the computational model contains not the whole core, but one fuel assembly with a height of 100 cm. The fuel volume in such a model is $V_{model} \approx 1,506 \cdot 10^4 \text{ cm}^3$.

Therefore, to obtain normalization per one fission in the computational model, the ratio of the core and model volumes is used:

$$T_{model} = \frac{R_{Rh}}{F_{AZ}} \cdot \frac{V_{AZ}}{V_{model}} \approx 10^{-6} \frac{\text{reactions in Rh}}{\text{neutron in the model}}. \quad (11)$$

The obtained value should be regarded as a lower estimate of the detector signal. The simplified calculation uses the average fission density in the fuel, whereas the detector is located in the central channel, where the thermal neutron flux can be higher than in the fuel. In addition, Monte Carlo models use an infinite geometry with $k_{\text{eff}} > 1$, which also increases the neutron field near the detector. Therefore, this estimate is used only to assess the order of magnitude of the results.

3.4 Estimation of the Neutron Collection Region

Before performing the main stage of calculations, it is necessary to estimate the size of the region around the detector from which neutrons make a significant contribution to the formation of its signal. This makes it possible to justify the dimensions of the computational model and to reduce the influence of boundary conditions on the result.

The size of the collection region depends on the absorptive properties of the medium. The smaller the macroscopic absorption cross section, the greater the distance a neutron can travel before interacting with the emitter material. Therefore, an upper estimate of the collection region should be obtained for the case of

the lowest absorption, namely for a medium with the minimum coolant density $\rho = 0,6 \text{ g/cm}^3$ and zero boric acid concentration $C_B = 0 \text{ g/kg}$.

The characteristic size of the region from which the SPND signal is formed is determined by neutron slowing down and subsequent diffusion processes. To estimate this size, the slowing-down length and diffusion length can be used. The slowing-down length characterizes the root-mean-square neutron displacement prior to thermalization and is approximately $\lambda_s \approx 50 \text{ cm}$ for light water.

The diffusion length of thermal neutrons in light water is of the order of $L \approx 3 \text{ cm}$.

The characteristic radius of the collection region can then be estimated as the sum of the slowing-down length and several diffusion lengths: $R \sim \lambda_s + 4L \approx 62 \text{ cm}$.

Thus, the main contribution to the SPND signal is formed by neutrons arriving from a region with a radius on the order of 25 cm around the detector. This value is used as an initial estimate of the minimum size of the computational region. The final choice of model dimensions is made by numerically comparing the detector signal as the simulation region is successively enlarged.

3.4.1 Estimation of the Neutron Collection Region Along the Model Height

The minimum height of the computational model is determined by the geometric dimensions of the detector and is equal to its length, $l = 25 \text{ cm}$. To study the contribution of more distant regions of the medium, the model height was successively increased. The expansion was performed symmetrically with respect to the detector center by adding coolant layers above and below.

The total model height was defined as

$$H = n \cdot l, \quad n = 1, 2, 3, \dots \quad (12)$$

This approach makes it possible to determine the height beyond which the contribution of additional regions to the SPND signal becomes statistically insignificant. This effectively defines the boundary of the neutron collection region along the model height.

The influence of reflective boundary conditions on the SPND signal was evaluated separately. A reflective boundary condition assumes that a neutron reaching the boundary of the computational region is reflected back into the model without energy loss and without changes in its other characteristics. Physically, this corresponds to the assumption of a symmetric continuation of the medium beyond the computational region.

For a correct comparison of models with different heights, the number of simulated particle histories was selected so that the relative statistical uncertainty of the signal remained approximately the same in all cases. Since the analysis was performed under conditions of a constant neutron flux density, the obtained values were additionally scaled according to the model height. The calculation results are presented in Table 1.

H , cm	without “mirrors”	“mirrors” at the ends	“mirrors” on all boundaries
25	$8,1 \pm 0,0$	$13,9 \pm 0,3$	$28,7 \pm 0,3$
50	$12,9 \pm 0,1$	$14,3 \pm 0,2$	$28,0 \pm 0,1$
75	$13,3 \pm 0,1$	$14,9 \pm 0,1$	$29,6 \pm 0,4$
100	$13,4 \pm 0,2$	$14,3 \pm 0,2$	$24,9 \pm 0,1$
125	$13,4 \pm 0,2$	$14,1 \pm 0,1$	$25,0 \pm 0,5$
150	$13,4 \pm 0,3$	$13,6 \pm 0,2$	$23,1 \pm 0,2$
175	$13,4 \pm 0,3$	$13,4 \pm 0,1$	$22,4 \pm 0,2$

Табл. 1: Dependence of the detector signal on the model height H for different boundary conditions per 10^5 generated neutrons

As follows from the results, as the model height increases, the SPND signal grows only up to a certain value of H , after which the increment becomes comparable

with the statistical uncertainty of the calculation. For models with $H \geq 3l$, a further increase in the computational region does not lead to a significant change in the signal. This indicates saturation of the contribution from remote regions of the medium; therefore, using a model with $H = 3l$ is sufficient for further calculations.

In addition, the results show that a significant part of the signal is formed in the region directly adjacent to the detector plane. In particular, the contribution from the region with height $H = l$ is about 60% of the signal obtained for the extended model. This confirms the local character of SPND response formation.

The use of reflective boundary conditions leads to an increase in the calculated signal. This is because neutrons that reach the model boundary are not lost, but return to the computational region and can interact again with the emitter material.

3.4.2 Estimation of the Neutron Collection Region in the Radial Direction

The neutron collection region in the radial direction was estimated by analyzing the contribution of fuel elements located at different distances from the detector. Radial enlargement of the computational region requires a more complex geometry and a substantial increase in computational time; therefore, the usefulness of such an expansion is determined by the ratio between the expected contribution from distant regions and the statistical uncertainty of the calculation.

For the model of one fuel assembly, the relative statistical uncertainty of the Monte Carlo calculation is approximately 0,5% at a computation time of about 10 h. At the same time, the contribution of peripheral fuel rods to the formation of the detector signal is smaller than this uncertainty (Fig. 8). Therefore, a further detailed increase in the model radius by adding distant fuel rods does not lead to a statistically significant change in the result and is not justified.

A comparison of models with and without reflective boundary conditions shows

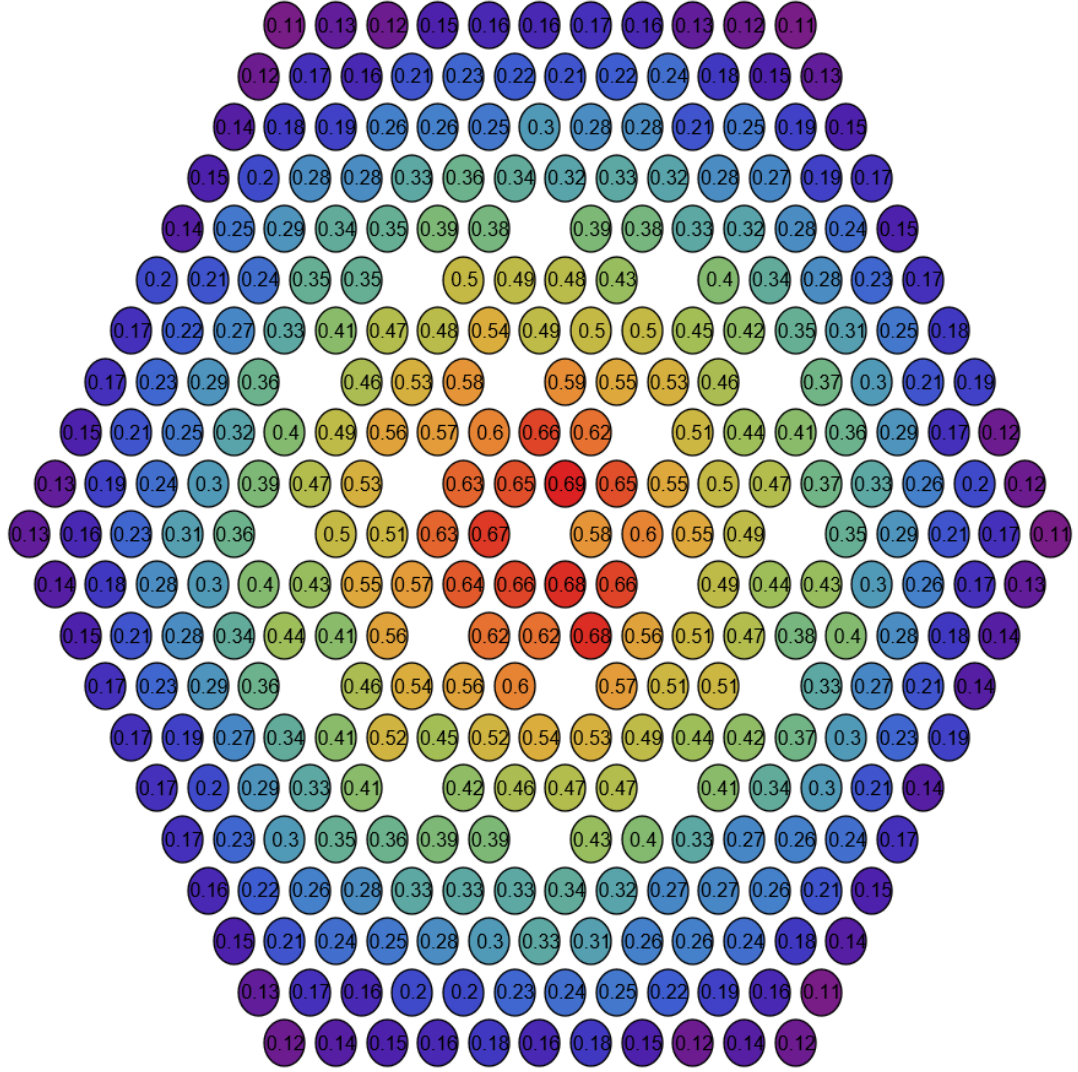


Рис. 8: Estimated contribution of fuel rods to the SPND signal depending on their position relative to the detector

a difference in the signal by approximately a factor of two. This indicates a contribution on the order of one half of the total signal from neutrons entering the detector region from neighboring regions of the core. This result confirms the importance of correctly accounting for the radial surroundings of the SPND or applying appropriate boundary conditions.

Thus, when choosing the radial size of the model, it is necessary to balance the accuracy of describing the surrounding medium against computational costs. For further calculations, it is reasonable to use a model that explicitly describes the nearest surroundings of the detector, while the influence of more distant regions can be accounted for through boundary conditions and appropriate normalization

of the results.

3.5 Influence of Boric Acid Concentration in the Coolant on the SPND Signal

The first stage in studying the influence of reactor core parameters on the SPND signal is the analysis of changes in boric acid concentration in the coolant. Boric acid is an effective neutron absorber; therefore, its concentration affects the local neutron field in the detector region and, correspondingly, the number of (n, γ) reactions in the rhodium emitter.

At the beginning of the fuel campaign, the boric acid concentration in the coolant can reach approximately 8 g/kg, whereas in borated water reserve tanks its value is 40 g/kg. Taking this into account, the boric acid concentration in the calculations was varied in the range from 0 to 10 g/kg. This range covers operationally significant values with a small margin.

The calculation results for the dependence of the SPND signal on the boric acid concentration in the coolant are shown in Fig. 9, while the corresponding data table is presented in Appendix A.

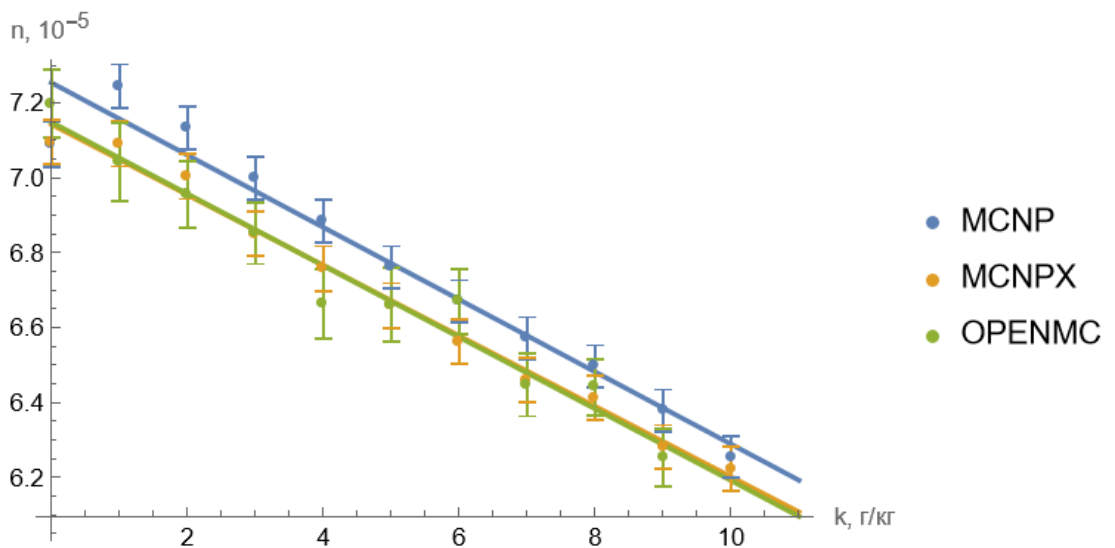


Рис. 9: Dependence of the SPND signal per 10^5 generated neutrons on the boric acid concentration in the coolant

The obtained data were approximated by a first-order polynomial using the least-squares method. The approximation parameters are given in Table 2.

Code	a	b
MCNP	7.254 ± 0.039	-0.097 ± 0.007
MCNPX	7.143 ± 0.018	-0.094 ± 0.003
OpenMC	7.149 ± 0.035	-0.096 ± 0.006

Табл. 2: Approximation parameters for the dependence of the SPND signal on boric acid concentration per 10^5 generated neutrons

3.6 Influence of Coolant Density on the SPND Signal

The next stage of the study is an analysis of the influence of coolant density on the self-powered neutron detector signal. In a VVER reactor, changes in coolant density are primarily associated with the temperature gradient along the core height. A change in water density affects neutron slowing down and absorption processes and, consequently, modifies the local neutron field in the SPND region.

To determine the coolant density, data from the National Institute of Standards and Technology (NIST) [12] were used. Water density values were determined as a function of temperature at a pressure of 15,6 MPa. The obtained data are listed in Table 3 and were used to specify the coolant density in the computational model.

Temperature, °C	Density, kg/m ³
290.00	746.37
295.00	736.76
300.00	726.70
305.00	716.14
310.00	704.98
315.00	693.14
320.00	680.49
325.00	666.83

Табл. 3: Coolant density as a function of temperature at a pressure of 15,6 MPa

The calculation results for the dependence of the SPND signal on coolant density are shown in Fig. 10, while the corresponding data table is provided in Appendix B.

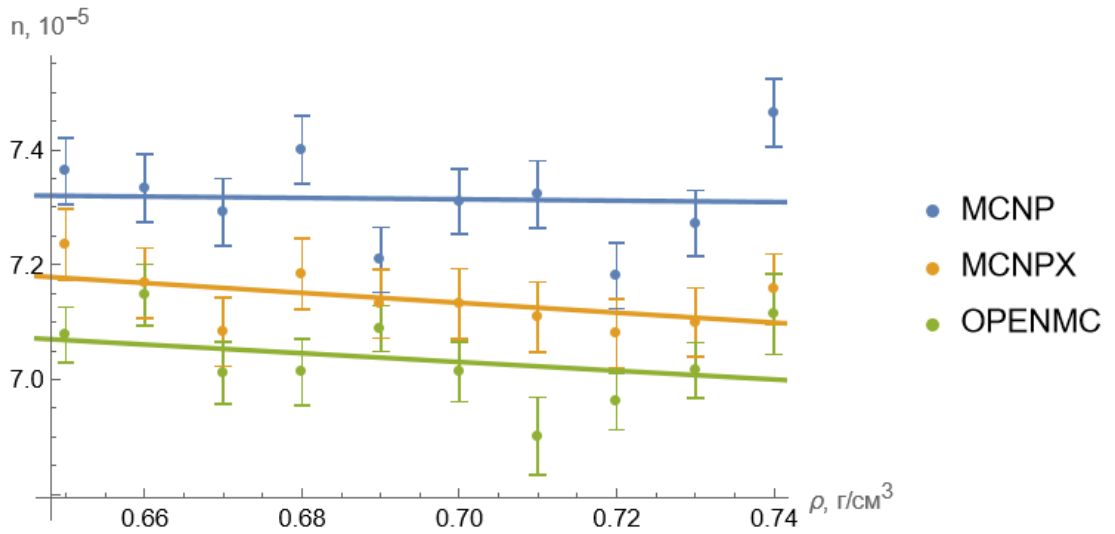


Рис. 10: Dependence of the SPND signal on coolant density per 10^5 generated neutrons

The obtained data were approximated by a first-order polynomial using the least-squares method. The approximation parameters are shown in Table 4. Within the considered density range, the dependence of the detector signal on coolant density is linear.

Code	a	b
MCNP	7.40 ± 0.68	-0.12 ± 0.98
MCNPX	7.74 ± 0.34	-0.86 ± 0.48
OpenMC	7.56 ± 0.57	-0.76 ± 0.82

Табл. 4: Approximation parameters for the dependence of the SPND signal on coolant density per 10^5 generated neutrons

3.7 Influence of Fuel Burnup on the Detector Signal

The next stage of the study is to account for the influence of nuclear fuel burnup on the self-powered neutron detector signal. To obtain the fuel isotopic composition during irradiation, the *depletion and transmutation* module of the OpenMC software package [13] is used.

The burnup calculations employ the Predictor-Corrector CE/CM (Constant Extrapolation / Constant Midpoint) algorithm, in which reaction rates over a time step are refined by repeating the neutron transport calculation for an intermediate isotopic composition [14]. At the predictor stage, the neutron flux density and reaction rates are assumed constant during the time step, after which the material isotopic composition at the midpoint of the interval is estimated. At the corrector stage, the transport calculation is repeated for this intermediate composition, and the obtained reaction rates are used for the final solution of the burnup equations.

This approach makes it possible to account more correctly for changes in the neutron field caused by the accumulation of fission products, formation of transuranic isotopes, and depletion of fissile nuclides in the fuel. Since changes in the fuel isotopic composition affect the neutron spectrum and flux density near the detector, they also change the number of (n, γ) reactions in the rhodium emitter and, consequently, the SPND output signal.

To verify the correctness of the obtained isotopic composition, the calculation

results were additionally compared with data from the SFCOMPO 2.0 database (Spent Fuel Isotopic Composition)[15,16,17], which contains experimental data on spent nuclear fuel composition.

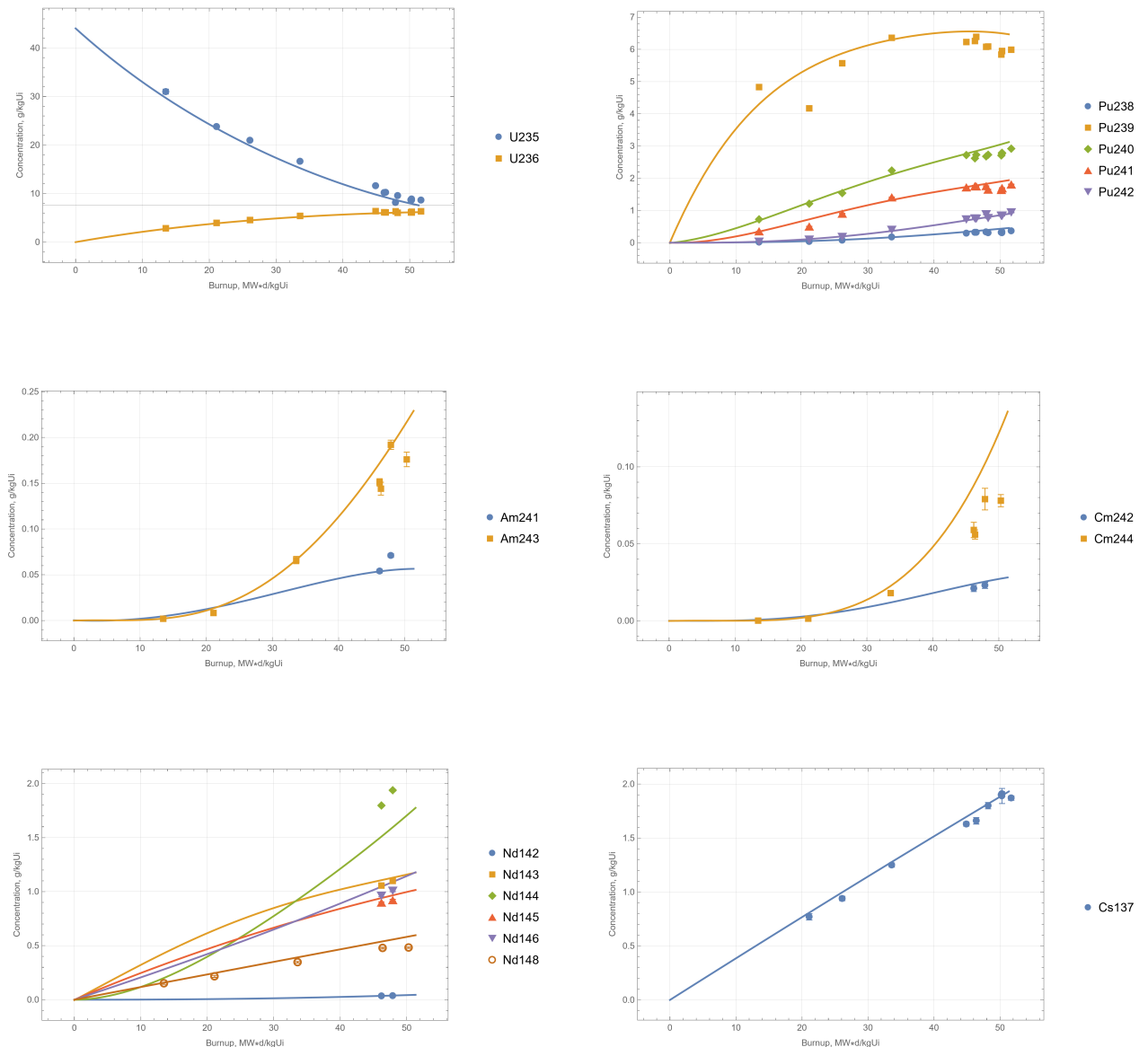


Рис. 11: Dependence of isotope concentrations on fuel burnup. Lines represent approximations of the calculation results, while markers denote experimental data. Concentrations are given in g/kg of initial uranium.

To quantitatively assess the agreement between calculated and experimental data, the relative root-mean-square deviation was used. For most isotopes, it does not exceed 15%, indicating satisfactory accuracy of the burnup model. The largest deviations are observed for minor actinides (^{243}Am , ^{244}Cm), which is associated

with the high sensitivity of their accumulation to neutron spectrum parameters and nuclear data uncertainties. The relative deviation values for the investigated isotopes are presented in Appendix C.

The calculation results for the dependence of the SPND signal on fuel burnup are shown in Fig. 12. The obtained dependence was approximated by a first-order polynomial, and the corresponding approximation parameters are provided in the table. The table containing the calculated data is given in Appendix D.

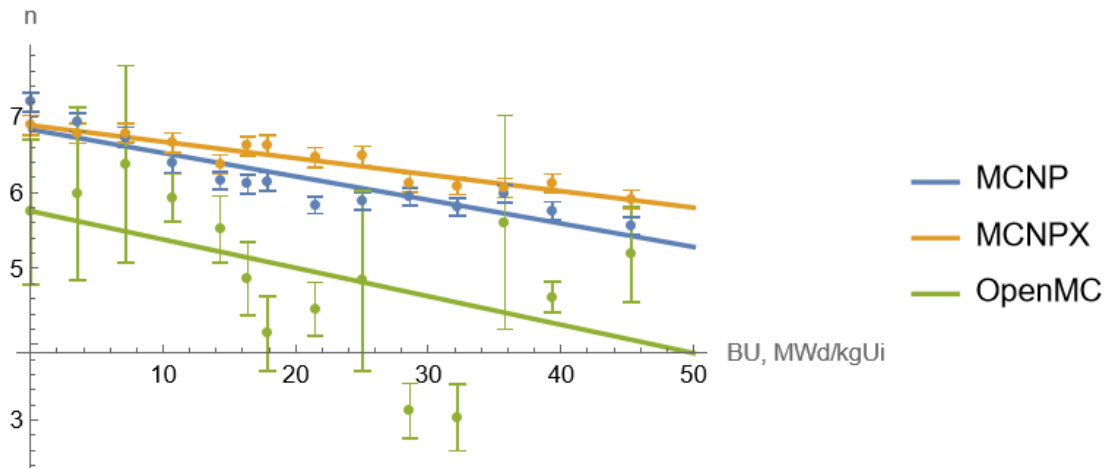


Рис. 12: Dependence of the SPND signal on fuel burnup per 10^5 generated neutrons

Code	a	b
MCNP	6.8 ± 0.1	-0.031 ± 0.004
MCNPX	6.89 ± 0.05	-0.022 ± 0.002
OpenMC	5.8 ± 0.5	-0.04 ± 0.02

Табл. 5: Approximation parameters for the dependence of the SPND signal on fuel burnup per 10^5 generated neutrons

4 Generalization of the Obtained Dependencies of the SPND Signal

At the previous stages, the influence of boric acid concentration in the coolant, coolant density, and nuclear fuel burnup on the self-powered neutron detector signal was investigated separately. For each of the considered parameters, the obtained numerical results were approximated by a first-order polynomial.

To estimate the simultaneous influence of all three factors, the individual dependencies were combined into a generalized model. It was assumed that the relative change in the signal caused by each parameter can be taken into account independently. In this approximation, the total dependence of the SPND signal can be represented as

$$n(k, \rho, B) = n_0 \frac{n_k(k)}{n_k(k_0)} \frac{n_\rho(\rho)}{n_\rho(\rho_0)} \frac{n_B(B)}{n_B(B_0)}, \quad (13)$$

where $n_k(k)$, $n_\rho(\rho)$, and $n_B(B)$ are the approximated dependencies of the detector signal on boric acid concentration k , coolant density ρ , and fuel burnup B , respectively; k_0 , ρ_0 , and B_0 are the reference parameter values; and n_0 is the signal value at the reference point.

The following reference values were used in this work:

$$k_0 = 0 \text{ g/kg}, \quad \rho_0 = 0,70 \text{ g/cm}^3, \quad B_0 = 0 \text{ MW} \cdot \text{day/kgU}. \quad (14)$$

Based on the approximated dependencies obtained in the MCNP, MCNPX, and OpenMC calculations, the corresponding total expressions for the SPND signal were constructed.

The obtained total dependencies make it possible to estimate the change in the SPND signal under simultaneous variation of the main reactor core parameters considered in this work. The constructed heat maps show a general tendency of the detector signal to decrease with increasing boric acid concentration and fuel

$$\text{MCNP} \quad n(k, \rho, B) = 0.02(6.83 - 0.03B)(7.25 - 0.10k)(7.40 - 0.12\rho)$$

$$\text{MCNPX} \quad n(k, \rho, B) = 0.02(6.89 - 0.02B)(7.14 - 0.09k)(7.74 - 0.86\rho)$$

$$\text{OpenMC} \quad n(k, \rho, B) = 0.02(5.76 - 0.04B)(7.15 - 0.10k)(7.56 - 0.76\rho)$$

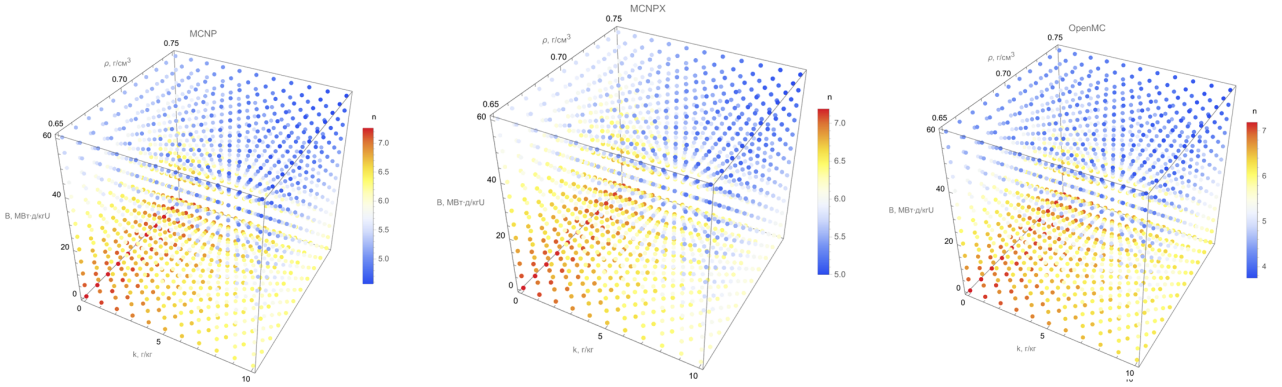


Рис. 13: Three-dimensional heat maps of the total dependence of the SPND signal on boric acid concentration, coolant density, and fuel burnup: MCNP, MCNPX, and OpenMC, respectively.

burnup. The influence of coolant density is also included in the generalized model; however, its character and magnitude differ somewhat among the results obtained with MCNP, MCNPX, and OpenMC.

5 Conclusions

This work investigated the influence of reactor core parameters on the signal of a self-powered neutron detector with a rhodium emitter.

A computational model of the FA-W assembly was constructed, including the self-powered neutron detector, fuel elements, and coolant, which makes it possible to account for local features of neutron field formation near the detector.

To account for changes in the fuel isotopic composition, burnup modeling of the FA-W assembly was performed.

A methodology was developed for accounting for the influence of the main operating parameters of the reactor core on the SPND signal, namely nuclear fuel burnup, boric acid concentration in the coolant, and coolant density. For each of these factors, both individual and combined dependencies of the detector signal were obtained.

It was established that changes in the considered parameters may lead to deviations of the SPND signal at a level of about 20%. Such an influence is significant; therefore, accounting for fuel burnup, boric acid concentration, and coolant density is justified and advisable in practical modeling and interpretation of self-powered neutron detector signals.

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7 Appendices

7.1 Appendix A. Dependence of Signal N on Changes in Boric Acid Concentration in the Coolant

k , g/kg	MCNP	MCNPX	OPENMC
0	7.09 ± 0.06	7.10 ± 0.06	7.20 ± 0.09
1	7.24 ± 0.06	7.09 ± 0.06	7.04 ± 0.10
2	7.13 ± 0.06	7.00 ± 0.06	6.96 ± 0.09
3	7.00 ± 0.06	6.85 ± 0.06	6.85 ± 0.08
4	6.88 ± 0.06	6.76 ± 0.06	6.66 ± 0.09
5	6.76 ± 0.06	6.66 ± 0.06	6.66 ± 0.10
6	6.67 ± 0.06	6.56 ± 0.06	6.67 ± 0.09
7	6.57 ± 0.06	6.46 ± 0.06	6.45 ± 0.08
8	6.50 ± 0.06	6.41 ± 0.06	6.44 ± 0.08
9	6.38 ± 0.06	6.28 ± 0.06	6.25 ± 0.08
10	6.25 ± 0.06	6.22 ± 0.06	6.22 ± 0.06

Табл. 6: Obtained data for the dependence of the detector signal on changes in boric acid concentration in the coolant per 10^5 generated neutrons

7.2 Appendix B. Dependence of Signal N on Coolant Density

ρ , g/cm ³	MCNP	MCNPX	OpenMC
0.65	7.36324 ± 0.0581696	7.23512 ± 0.062222	7.07835 ± 0.0481632
0.66	7.33318 ± 0.0586654	7.16865 ± 0.0609335	7.14744 ± 0.0536687
0.67	7.2914 ± 0.0583312	7.08332 ± 0.0602082	7.01132 ± 0.0542076
0.68	7.40026 ± 0.0599421	7.18414 ± 0.0617836	7.0131 ± 0.0585562
0.69	7.20861 ± 0.056948	7.13203 ± 0.0599091	7.08893 ± 0.0394691
0.70	7.31059 ± 0.0570226	7.13241 ± 0.0606255	7.0136 ± 0.0520845
0.71	7.32204 ± 0.0585763	7.10926 ± 0.0611396	6.90158 ± 0.068121
0.72	7.18137 ± 0.057451	7.0805 ± 0.0601843	6.96153 ± 0.0491808
0.73	7.27173 ± 0.0574467	7.09964 ± 0.0603469	7.0166 ± 0.0477624
0.74	7.46493 ± 0.0597194	7.1573 ± 0.0615528	7.11386 ± 0.0697714

Табл. 7: Dependence of signal N on coolant density ρ per 10^5 generated neutrons

7.3 Appendix C. OpenMC Calculation Results

Isotope	RMS, g/kgUi	RMS, %
Am241	0.011	15.90
Am243	0.021	27.07
Cm242	0.003	12.61
Cm244	0.026	53.17
Cs137	0.050	3.64
Nd142	0.002	4.38
Nd143	0.041	3.87
Nd144	0.317	16.88
Nd145	0.043	4.70
Nd146	0.084	8.53
Nd148	0.060	14.52
Pu238	0.072	24.42
Pu239	0.544	11.03
Pu240	0.200	7.60
Pu241	0.147	14.68
Pu242	0.035	9.54
U235	1.056	9.27
U236	0.178	3.07

Табл. 8: Root-mean-square deviations between the OpenMC calculation results and the experimental data from the SFCOMPO database

7.4 Appendix D. Dependence of Signal N on Fuel Burnup

B , MW·day/kg $_{U_i}$	MCNP	MCNPX	OpenMC
0	7.19 ± 0.13	6.89 ± 0.13	5.75 ± 0.95
3.58	6.93 ± 0.12	6.78 ± 0.13	5.99 ± 1.14
7.16	6.73 ± 0.14	6.78 ± 0.13	6.37 ± 1.30
10.73	6.38 ± 0.12	6.65 ± 0.13	5.93 ± 0.31
14.31	6.16 ± 0.12	6.37 ± 0.12	5.52 ± 0.44
16.39	6.11 ± 0.12	6.61 ± 0.13	4.86 ± 0.48
17.89	6.15 ± 0.12	6.63 ± 0.13	4.14 ± 0.49
21.47	5.83 ± 0.11	6.46 ± 0.13	4.46 ± 0.36
25.05	5.89 ± 0.11	6.49 ± 0.13	4.84 ± 1.19
28.63	5.95 ± 0.12	6.12 ± 0.13	3.12 ± 0.36
32.20	5.81 ± 0.11	6.09 ± 0.12	3.03 ± 0.44
35.78	5.98 ± 0.12	6.06 ± 0.13	5.60 ± 1.41
39.36	5.76 ± 0.12	6.12 ± 0.12	4.62 ± 0.20
45.34	5.56 ± 0.12	5.91 ± 0.12	5.18 ± 0.62

Табл. 9: Dependence of the detector signal on fuel burnup
per 10^5 generated neutrons

7.5 Appendix E. Input File Listing

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3  3  -6.55    -14 13  imp:n=1  u=1
4  4  -0.75    14  imp:n=1  u=1
5  4  -0.75  -21  imp:n=1  u=2
6  3  -6.55    -22 21  imp:n=1  u=2
7  4  -0.75    22  imp:n=1  u=2
20 6  -12.41    3 -4 -111 imp:n=1  u=3
21 7  -3.99    #20 -112 imp:n=1  u=3
22 3  -7.8     112 -113 imp:n=1  u=3
69 2  -0.000178 113 -180 imp:n=1  u=3
71 3  -7.8     180 -181 imp:n=1  u=3
72 4  -0.75  181 -21  imp:n=1  u=3
73 3  -6.55    21 -22  imp:n=1  u=3
74 4  -0.75  22  imp:n=1  u=3
80 4  -0.75  -101 -102 -103 -104 -105 -106 lat=2 fill=-11:11
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0 0 0 0 0 0 0 0 0 0 0 4 1 1 1 1 1 1 1 1 1 1 4
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```

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```

imp:n=1 u=4

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u=0

```

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83 0 (201:202:203:204:205:206:-1:2) imp:n=0
u=0

```

```

*1 pz -25
*2 pz 25
3 pz -12.5
4 pz 12.5
12 cz 0.392
13 cz 0.4
14 cz 0.457
21 cz 0.55
22 cz 0.63
111 cz 0.024

```

112	cz	0.05			
113	cz	0.065			
180	cz	0.35			
181	cz	0.39			
101	px	0.6375			
102	p	-1	0	0	0.6375
103	p	0.5	0.8660254	0	0.6375
104	p	-0.5	-0.8660254	0	0.6375
105	p	-0.5	0.8660254	0	0.6375
106	p	0.5	-0.8660254	0	0.6375
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*202	p	-0.8660254	-0.5	0	11.8
*203	py	11.8			
*204	p	0	-1	0	11.8
*205	p	-0.8660254	0.5	0	11.8
*206	p	0.8660254	-0.5	0	11.8

M1

8016.11 c -1.18498000E-01
92235.11 c -2.08951094E-02
92238.11 c -8.60606891E-01

M2 2004.11 c 1

M3 40090.11 c -97.3005
41093.11 c -1
50112.11 c -0.0121
50114.11 c -0.0083
50115.11 c -0.0043
50116.11 c -0.1818

	50117.11c	−0.0960
	50118.11c	−0.3028
	50119.11c	−0.1074
	50120.11c	−0.4073
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	50124.11c	−0.0724
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	26056.11c	−0.3395
	26057.11c	−0.0078
	26058.11c	−0.0010
	8016.11c	−0.06
	6012.11c	−0.00742
	6013.11c	−0.00008
	14028.11c	−0.00692
	14029.11c	−0.00035
	14030.11c	−0.00023
	7014.11c	−0.0045
M4	1001.11c	−0.11127058
	8016.11c	−0.88698119
	5010.11c	−0.00034790
	5011.11c	−0.00140033
MT4 LWTR.04T		
M6	45103.11c	1
M7	13027.11c	2
	8016.11c	3
NONU		
TOTNU		
NPS	1E8	

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EXT=D3

ERG=D4

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82: 80(1 -9 0):1
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 Fc14 Emitter - (n,gamma)
 Fm14 (-1 6 102)
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